MILITARY SPECIFICATION
COMPONENTS, ROCKET PROPULSION FLUID SYSTEM, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers the requirements and instructions for preparation of the manufacturer's model specification for all components of a rocket propulsion system which control the pressure or flow of ullage, gas, pressurizing gas, propellants, combustion products, or hydraulic fluid. Specifically excluded from this specification are tankage, turbo machinery, injectors, thrust chambers, nozzles, and gas generators.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-S-4040	Solenoid, Electrical, General Specification For
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-6021	Castings, Classification and Inspection of
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-P-6906	Plates, Identification
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification For

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MIL-W-8160	Wiring, Guided Missile, Installation of, General Specification For
MIL-E-8189	Electronic Equipment Guided Missiles, General Specification For
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter; General Specification For
MIL-M-9950	Missile Component, Liquid Oxygen Nitrogen Gaseous Oxygen, Instrument Air, Helium and Fuel Handling Systems, Cleaning and Packaging for Delivery
MIL-N-25027	Nut, Self-Locking, 250 Deg. F, 450 Deg. F, and 800 Deg. F, 125 KSI FTU, 60 KSI FTU, and 30 KSI FTU
MIL-E-25366	Electric and Electronic Equipment and Systems. Guided Missiles, Installation of, General Specification For
MIL-H-25475	Hydraulic Systems, Missile, Design, Installation Tests, and Data Requirements, General Requirements For
MIL-P-27407	Propellant, Pressurizing Agent, Helium
MIL-P-27409	Propellant Feed System, Rocket Propulsion, General Specification For

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-446	Environment for Electronic Parts, Tubes and Solid State Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-785	Reliability Program for Systems, Subsystems, and Equipment

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MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MS 33540	Safety Wiring, General Practices For
MS 33588	Nuts and Plate Nuts, Self-Locking, Aircraft Design and Usage Limitations of

PUBLICATIONS

Air Force - Navy Aeronautical Bulletins

No. 343	Specifications and Standards Applicable to Aircraft Engines and Propellers, Use of
No. 438	Age Controls of Age-Sensitive Elastomeric Items
No. 445	Engineering Changes to Weapons, Systems, Equipments and Facilities

3. REQUIREMENTS

3.1 General.

3.1.1 Model specifications. A model specification conforming to the Appendix specified herein shall be prepared by the contractor or manufacturer for approval by the procuring activity (6.3.8).

3.1.2 Pre-production. Prior to initiating production of components for use in flight systems components shall have successfully completed the Qualification tests (4.2.2).

3.1.3 Changes. The contractor shall notify the procuring activity of any proposed design changes to the component as defined in the model specification. Following successful completion of the Qualification tests, no change shall be made except when such changes are made in accordance with ANA Bulletin No. 445 or procurement activity requirements as applicable. Approval of changes does not relieve the contractor from full responsibility for the effects of the changes on the component performance.

3.1.4 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2 Materials and processes. The materials and processes used in the manufacturer of components shall conform to the applicable specifications, in accordance with paragraph entitled "Material Specifications" of ANA Bulletin No. 343 and shall be selected from the list of materials and processes approved by the procuring activity for the specific application.

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3.2.1 Metals. Metals shall be compatible with fluids (3.4.8), fluid vapors, fluid decomposition products, products of combustion where applicable, and with environments to which subjected. In addition, they shall not deteriorate in normal usage and storage. The use of dissimilar metals, especially brass, copper, or steel in contact with aluminum alloy, shall be avoided where practicable. Protection of dissimilar metal combinations shall be in accordance with MIL-STD-454 Requirement 16.

3.2.2 Rubbers, plastics, and elastomers. Rubbers, plastics, and elastomers shall be compatible with fluids (3.4.8), fluid vapors, fluid decomposition products, products of combustion where applicable, and with environments to which subjected. Age control of synthetic rubber parts shall be in accordance with ANA Bulletin No. 438.

3.2.3 Lubricants. Propulsion system components shall not require added lubrication during their normal service life (6.3.11). All component wear surfaces in contact with system fluids shall not require additional lubrication other than the system fluid itself. Lubricants shall be compatible with fluids, fluid vapors, fluid decomposition products, products of combustion where applicable, and with environments to which subjected.

3.2.4 Finishes. All finishes, surface treatment, plating, coatings, or paints shall be specified in the model specification.

3.3 Performance characteristics. The performance characteristics, including the intended application and the fluid system interactions, shall be specified in the model specification. The following shall be included as a minimum.

3.3.1 Operating attitude. Components shall be capable of meeting the performance characteristics specified in the model specification when operated in any attitude the components may be exposed to during their operating life (6.3.6). The operating attitudes shall be specified in the model specification.

3.3.2 Life characteristics. The component operating and storage life (6.3.12) characteristics shall be specified in the model specification.

3.3.2.1 Storage life. All rocket fluid system components shall have a minimum required storage life (non-operating) of 5 years. The internal and external environmental extremes to be encountered during storage shall be specified in the model specification. Where the component is in contact with propellants or actuating fluids during storage, it shall not degrade these fluids.

3.3.2.2 Operating life. The operating life shall be specified in the model specification, and shall include acceptance testing, system checkout tests, and propulsion system operation.

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3.3.3 Leakage. The maximum component leakage (internal and external) and measurement technique shall be specified in the model specification. Where fuel and oxidizer are controlled in a single component, seal leakage shall be ducted such that mixing of fuel and oxidizer within the component is precluded.

3.3.4 Flow characteristics. Component flow characterictics shall be specified in the model specification and shall include flow rate versus pressure drop at given temperature for various percentages of maximum flow, or flow rate versus percentage maximum flow position of the control element at given temperature and pressure drop.

3.3.5 Actuation. The method of actuation, response characteristics from application of actuation signal to fully open and closed conditions, first order time constant, repeatability of the response characteristics, time from valve first movement to fully open or closed position, and the control mode, i.e. digital, proportional, etc., shall be specified in the model specification.

3.3.6 Power. The power range (electrical, hydraulic, pneumatic or other) for which the component must meet the actuation requirements, over the operating temperature range; the tolerances and characteristics of the power available to the component, and the maximum power to which the component may be subjected shall be specified in the model specification.

3.3.7 Stability characteristics. Stability characteristics to include overshoot, undershoot, damping, and dead band conditions shall be specified in the model specification.

3.3.8 Filtration. Nominal (6.3.5) and absolute filtration (6.3.1) along with the dirt holding capacity at rated flow and pressure drop of all filter elements shall be specified in the model specification.

3.3.9 Draining. Drainage provisions shall be made so that propellants cannot be trapped during unloading (ground only). All components for cryo genic service shall have provisions for relieving pressure from any trapped compartments. Maximum spillage which may occur in disconnect operations shall be specified.

3.4 Design. Components shall be designed to provide high inherent reliability (6.3.10), permit use of proven manufacturing processes and positive inspection at each level of manufacture, have low susceptability to assembly errors, minimize variation of performance caused by manufacturing tolerances, and minimize weight while achieving the performance characteristics (3.3) specified in the model specification, when subjected to the expected operating environment.

3.4.1 Misalignment. Maximum allowable interface loads on the component caused by axial, lateral, and angular misalignments in the tubing or the support structure under the most severe environmental condition including any influences from deflection or misalignment accomodation features shall be specified in the model specification.

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3.4.2 Standards. MS or AN standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their standard part numbers. The use of nonstandard parts will be acceptable only when standard parts have been determined to be unsuitable. MS and AND design standards shall conform to applicable standards in accordance with paragraph entitled "Utility Parts Standards" in ANA Bulletin No. 343.

3.4.3 Special tools. The design of system components shall be such that disassembly and replacement of parts required in operational maintenance shall be accomplished without the use of special tools which are unique to the component.

3.4.4 Pressure. The working (6.3.13) proof (6.3.9), and burst pressure (6.3.2) shall be specified in the model specification. The proof and burst pressures shall be at least 1.5 and 2 times the working pressure respectively and they shall be adjusted for the difference in test temperature and maximum working temperature to account for the reduced physical properties.

3.4.5 Installation characteristics. Mounting details; overall dimensions, location, size and type of plumbing and electrical connections; servicing clearances, weight, center of gravity, and flow directions shall be specified in the model specification.

3.4.6 Environment. The environments the component will experience in flight, i.e. vibration, acceleration, temperature and altitude, as well as the non-operating environments such as shipping vibration and shock, the natural environments of the launch and test sites, and the launch environment for upper stage systems components shall be specified in the model specification.

3.4.7 Contamination. Contamination levels and quantities of contaminants at each particle size level, which the component must be capable of accepting without degradation in performance, shall be specified in the model specification. However, the levels and quantities of contaminants should not be less than the maximum allowed in the propellant specification.

3.4.8 Fluids. The chemical composition, allowable contamination levels (solids, liquids and gas), pressure and temperature ranges, and flow rates shall be specified for the actuating, working and test fluids. The contamination level specified shall be the maximum allowable for propulsion system operation (as opposed to the contamination levels called out in the fluid procurement specification).

3.5 Construction.

3.5.1 Threaded connections and attachments.

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3.5.1.1 Locking provisions. All threaded members such as nuts, screws, and bolts shall be provided with safety wiring or other acceptable means of locking. Any adjustments which may be provided shall be capable of being locked securely enough to prevent disturbance of setting under all service and test conditions. Self-locking nuts shall conform to MIL-N-25027 and shall be used in accordance with MS 33588. Safety wiring shall comply with the requirements of MS 33540. The use of staking and lock washers shall be avoided unless they can be shown to have superior characteristics to the more conventional methods and approval is obtained from the procuring activity.

3.5.1.2 Pipe threads. Pipe threads shall not be used on components.

3.5.1.3 Threads. Conventional straight screw threads shall conform to MIL-S-7742 or MIL-S-8879. Duplicate parts, differing only in thread form, shall not be permitted. Unless otherwise specified, threaded parts smaller than 0.164 inch diameter shall have threads in accordance with MIL-S-7742. When an allowance is required for application in elevated temperature, corrosive atmosphere, or other conditions which may cause thread seizure, this allowance shall be obtained by increasing the diameter of the internal threads.

3.5.2 Electrical devices. Components incorporating electrical devices shall conform to MIL-S-4040, MIL-W-8160 MIL-E-8189, MIL-E-25366, MIL STD 446, and each electrical connector used on a component shall incorporate a ground return terminal.

3.5.3 Hydraulic devices. Hydraulic actuation devices shall conform to the applicable requirements of Table I and paragraph 3.7 of MIL-H-25475. Hydraulic activation devices, which operate using the rocket engine's propellant as the working fluid, are excluded from the requirements of MIL-H-25475.

3.6 Reliability. The contractor shall include in the model specification the requirements for performing a detailed failure mode analysis. The failure mode analysis shall be used in designing the development and qualification tests and in reporting the test results. The model specification shall also include the manufacturing control and standards of MIL-STD-785. The contractor shall be responsible for establishing the reliability design reviews, reliability reporting procedures and reliability demonstration as specified in MIL-STD-785.

3.7 Interchangeability. All parts having the same manufacturer's part numbers shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number listed in MIL-D-1000. The use of non-interchangeable assemblies, i. e. lapped piston cylinder assemblies, shall be approved by the procuring activity and the assembly shall be given one part number.

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3.8 Accessibility. Where practicable, component parts requiring routine service checking, adjustment, or replacement shall be made readily accessible for servicing without tear-down of the units, removal of any major part, component, or accessory. Where periodic or field adjustment is required, positive locking of the adjustment shall be provided and provisions for sealing the adjustment shall be included to effectively prevent unauthorized adjustment.

3.9 Identifying products. Components shall be marked for identification in accordance with MIL-STD-130.

3.10 Nameplate. Nameplate conforming to MIL-P-6906 shall be securely attached to the component, or the information may be electrochemically etched, in a suitable location on the component. In addition to marking required by MIL-STD-130, the following shall be included.

a. Contractor's part number.

b. Serial number

c. Rating (i.e., working pressure, set point, etc.)

d. Date of acceptance.

3.11 Drawings and diagrams. Manufacturer's assembly and detail drawings shall conform to MIL-D-1000. A complete set of full size drawings and diagrams, together with the model specification, shall incorporate reduced size copies of the same drawings and diagrams in the body of the model specification and shall be prepared for the contractor by the manufacturer with the preproduction components.

3.12 Parts list. A manufacturer's parts list (6.3.7) shall be prepared for each unit and shall list, but not be limited to, such information as the unit nomenclature, manufacturer's part numbers, detail nomenclature, and numerical identification of each part.

3.13 Contamination control. Methods for cleaning and handling of each component and component element to assure that minimum cleanliness levels are compatible with propulsion system use shall be established. Minimum requirements for oxygen, nitrogen, air helium, and fuel systems shall be as specified in MIL-M-9950.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of tests. The tests to be performed shall be classified as follows:

a. Development tests (4.2.1)

b. Qualification tests (4.2.2)

c. Quality Conformance tests (4.2.3)

4.2.1 Development tests. Development tests consist of all tests performed prior to qualification testing on the preproduction components.

4.2.2 Qualification tests.

4.2.2.1 Sampling instructions. Unless otherwise specified, each qualification test sample shall consist of a set of components of each manufacturer's part number upon which approval is desired. The number of components in each set shall be those deemed necessary by the procuring activity to perform the applicable tests as indicated. One set of components shall have been tested by the manufacturer in accordance with this specification prior to being forwarded to the testing facility (each component shall be tested separately). The components shall be accompanied by one complete set of manufacturer's drawings, the component specification, the failure mode analysis, a parts list, and a complete test report showing the results of the contractor's tests. Samples shall be identified as required and forwarded to the testing facility designated in the contract.

4.2.2.2 Tests. The qualification of components shall consist of tests in Section 4.5 which are applicable to the particular type component. Any additional tests or more severe requirements may be specified. All qualification tests shall be specified in detail in the model specification.

4.2.3 Quality Conformance tests. Quality Conformance testing of propulsion fluid system components shall consist of inspection and performance tests required to assure components meet the standards of the model specification. Quality Conformance test procedure shall be reviewed following successful completion of qualification to determine its adequacy. The Quality Conformance test procedures shall be specified in detail in the model specification.

4.3 Failures and retest of test samples.

4.3.1 Development tests. Failures encountered during the development testing shall be reported on tests of the preproduction versions of the complete component. The design changes shall also be documented and the failure and change referenced to the failure mode analysis.

4.3.2 Qualification tests. Failures encountered during the Qualification testing shall be reported and the failure referenced against the failure mode analysis. If a component failure is encountered, a detailed

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failure analysis shall be performed to determine the cause of failure and necessary corrective action. Components may be reworked to comply with the corrective action. Following the corrective action the series of tests for the components shall be started again from the beginning. If a design change is required in the component or one of its elements, the design change shall be reported with the effects of the change on the failure mode analysis. A design change shall require restarting qualification for the component with components incorporating the design change unless waiver of retest is granted by the procuring activity.

4.3.3 Quality Conformance test. Failure of any component to pass the Quality Conformance test shall be cause for rejection of all components in the lot presented. An immediate investigation shall be made to determine the cause of performance degradation below the qualification standards. If a design change is required in the component or one of its elements, the modified design shall be requalified. If a variation in a manufacturing process caused the failure and closer control of the process or a different process is required to produce the required quality, all components of the lot will be reinspected to assure they are of acceptable quality. Inability to adequately inspect the parts to determine if they are of an acceptable quality shall cause their rejection and replacement.

4.4 Test conditions. Unless otherwise specified, the following test conditions shall apply during the tests performed in accordance with this specification.

4.4.1 Cleaning. Unless otherwise specified, the test system shall be cleaned and passivated to the approved standards established for the propulsion system. All test fluids shall be filtered to assure that the contamination level at the component does not exceed the propulsion system cleanliness specification.

4.4.2 Temperature and pressure. Unless otherwise specified, all tests shall be conducted at a temperature of 60° to 90°F (15.6° to 32.2°C) and at atmospheric pressure.

4.4.3 Attitude. Unless otherwise specified, all tests shall be conducted with the component installed in the normal operating attitude.

4.4.4 Test fluid. The test fluid for each test shall be specified in the model specification.

4.5 Tests and test methods. The contractor shall list and describe in detail the tests, test procedure, and test system design for the Qualification and Ouality Conformance tests. In addition, required instrumentation and allowable tolerances shall be specified in the model specification. All instrumentation upon which the accuracy of the test results depends shall be calibrated frequently enough to ensure the accuracy required by the model specification. Calibration records shall be maintained (4.1) and shall be made available to the procuring activity upon request. The instrumentation accuracy shall be subtracted from the allowable performance tolerance of the measurement to determine the as measured tolerance.

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4.5.1 Examination. The units shall be examined to assure compliance with the requirements of this specification. Such inspection shall include but shall not be limited to the requirements of identification marking, physical measurement, weight, continuity of required wiring, proper wiring, finish, freedom from damage, and maintenance of the required standard of workmanship. Radiographic plates for all castings, in accordance with MIL-C-6021, shall be presented with the unit at the time of acceptance inspection. These plates shall be identifiable to the particular unit.

4.5.2 Strength.

4.5.2.1 Proof pressure. The proof pressure shall be specified in the model specification, and shall be applied for not less than three (3) minutes. Where multizone components are being pressurized, the proof pressure sequency shall simulate system operation. Where the pressure ratio between zones will vary during opevation, two or more tests may be required to simulate maximum pressure differentials across various sections. Evidence of permanent deformation shall be cause for rejection.

4.5.2.2 Burst pressure. The component shall have the burst pressure as specified in the model specification, applied and maintained for a period of not less than three (3) minutes. The pressure shall be applied at all pressure ports (where the burst pressure varies between parts on a component the appropriate burst pressure will be applied to each part simultaneously.) The units shall not burst. The method of applying the pressure shall be specified in the model specification. Permanent set may occur but subsequent functioning of the units is not required. If practicable, pressure shall then be increased above the specified burst pressure until failure occurs. Failure pressure shall be recorded.

4.5.2.3 Surge pressure. The test system, procedure, and requirements of this test shall be specified in the model specification.

4.5.2.4 Mechanical load. With the component mounted in the normal manner and operating at maximum working pressure, it shall be subjected to the most severe tension, compression bending, and torsional loads that would be expected to occur in service. There shall be no evidence of failure, malfunction, permanent distortion or unsatisfactory leakage as a result of this test.

4.5.3 Leakage. Tests shall be conducted at pressures up to the working pressure to determine both internal and external leakage using the Helium, MIL-P-27407 or other fluid approved by the procuring activity (3.3.3).

4.5.3.1 Internal. The leakage shall be measured at five equal pressure increments from ambient to working pressure using an instrument with an accuracy and precision at least one order of magnitude lower than the allowable leakage for the component. For components with more than one outlet port, the total internal leakage will be measured. For components that must seal against flow in either direction, a reverse flow leakage test will also be performed.

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4.5.3.2 External. The external leakage shall be measured at five equal pressure increments from ambient to working pressure using an instrument with an accuracy and precision at least one order of magnitude lower than the allowable leakage for the component.

4.5.4 Functional.

4.5.4.1 Pressure drop. This test shall be applicable to all components whose function in the propulsion system is such as to affect the total pressure drop of the propulsion system. The pressure drop test shall be conducted over a range of temperature and flow rates that completely covers the design capacity of the unit, and sufficient data shall be recorded to permit the plotting of curves. The pressure drop graph shall show the pressure drop of the component using the test fluid, ΔP observed, and where a substitute fluid is used, the pressure drop of the component corrected to the propulsion system fluid.

4.5.4.2 Pressure control. Pressure control tests shall measure the stability characteristics for pressure control components. The start transients, steady state, lock up (6.3.4) and droop (6.3.3) characteristics or the crack, reseat, and flow stability shall be determined over a range of pressures and flow rates that completely covers the design capacity of the unit. The test system used for this test shall dynamically simulate the propulsion system installation. The instrumentation used for this test shall have a frequency response at least two times the minimum frequency response specified for the component in the model specification. The test procedure for this test and test configuration shall be specified in detail in the model specification.

4.5.4.3 Flow control. Flow control and stability characteristics of a component shall be measured by this test. The throttling characteristics, control input to flow characteristics, repeatibility of the control input to achieve a particular flow and stability of the component shall be determined for the complete range of pressures, flow rates and response rates within the design capacity of the component. The test system used for this test shall be dynamically equivalent to the propulsion system installation. This test shall be specified in detail in the model specification.

4.5.4.4 Response. In performing its design function, the response of the component shall be measured using instrumentation with a frequency response at least two times greater than the required response of the component. This test shall be conducted over the range of conditions which cover the design capacity of the unit. This test shall be specified in detail in the model specification.

4.5.4.5 Power requirements. The power requirements test shall be applicable to all components requiring an external source of power for their operation, whether this source be electrical, mechanical, manual, pneumatic, hydraulic, or other. This test shall be conducted over the

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complete range of operating flows, pressures, ambient and fluid temperatures, and other parameters affecting power as are specified. The minimum power required for satisfactory operation and the operating characteristics with maximum allowable power shall be measured.

4.5.5 Endurance. The endurance test shall be specified in the model specification. This test, as a minimum, shall equal the operating life requirement in 3.3.2.2, plus a margin of safety necessary to demonstrate the required reliability.

4.5.6 Electrical system.

4.5.6.1 Explosion-proof. The explosion-proof test shall be conducted in accordance with Procedure IV, MIL-E-5272.

4.5.6.2 High ambient air temperature. This test shall be conducted on all components which incorporate any electrical subcomponents and which are rated as "continuous duty" with the exception of pumps. This will include items which are solenoid operated. The test shall be conducted as follows:

The entire unit, dry, shall be placed into a chamber with the ambient air controlled within the limits of 160°F (71.1°C) and 170°F (76.7°C). It shall then be energized with the maximum allowable voltage; e. g., 30 volts DC, etc., applied to the power unit terminals, for a period of three hours. At approximately 10 minute intervals, the current flow, terminal voltage, ambient temperature, and power unit housing temperatures shall be recorded. The unit shall then be removed from the chamber and the functional test shall be conducted at room temperature with the minimum allowable voltage; e. g., 18 volts DC, etc., applied to the power unit terminals. The unit shall meet the performance requirements (3.5) for 30 cycles. Using data taken during the high temperature test, a curve shall be plotted showing the variation of coil temperature as determined by the resistance method with time.

4.5.6.3 Dielectric strength. This test shall be applicable to all components which include any electrical circuits. The applicable test voltage as specified in Table I shall be applied between circuits and between each circuit and the body of the component. Current flow in excess of 2 milli-amperes or breakdown of insulation shall constitute failure.

TABLE I

Test voltage

Nominal voltage	Component	Test voltage	Time
28V dc	Motor	600V or 500V	1 second 1 minute
28V dc	Other than motors	1,000V	1 minute
115/200V 400 cps ac	All	1,500V	1 minute

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4.5.6.4 Radio interference. Unless otherwise specified, all propulsion system components employing any electrical circuits shall conform to MIL-I-6181. The detailed test procedure for each particular component shall be specified in the model specification.

4.5.7 Fluid compatibility.

4.5.7.1 Internal. The compatibility of the component with the actuating and working fluids (3.4.8) shall be demonstrated at the maximum fluid pressure and at the maximum and minimum fluid temperature. The minimum test period at each condition shall be 48 hours. The component shall be operated once every six hours. Degradation of the fluid or reduction of the component performance characteristics below those specified (3.3) shall disqualify the component.

4.5.7.2 External. With the component mounted in its normal attitude, it shall be subjected to a spray of the propellant until all exposed surfaces are wetted. Then air dry the component for one day and inspect for detrimental corrosion or other evidence of incompatibility. Performance of this test is required for the fuels and oxidizers only.

4.5.8 Environment.

4.5.8.1 Ground handling and storage.

4.5.8.1.1 Mechanical shock. The shock test shall be conducted in accordance with MIL-STD-810. The procedure shall be specified in the model specification.

4.5.8.1.2 High temperature. High temperature test shall be in accordance with MIL-STD-810.

4.5.8.1.3 Low temperature. Low temperature test shall be in accordance with MIL-E-5272.

4.5.8.1.4 Temperature shock. Temperature shock test shall be in accordance with MIL-STD-810.

4.5.8.1.5 Humidity. With all ports suitably connected as in use, including electrical receptacle, test the unit in accordance with the humidity test in MIL-STD-810.

4.5.8.1.6 Salt fog. With all ports suitably connected as in use, the units shall be subjected to the salt spray test as outlined in MIL-E-5272.

4.5.8.1.7 Sand and dust. With all ports suitably connected as in use, the units shall be subjected to the sand and dust test in MIL-E-5272.

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4.5.8.2 Flight environment. Two types of flight environment tests shall be performed. The first type simulates flight environments encountered when the component is not operating. This condition is typical of upper stage components during boost. The second type simulates environments which are encountered when the component is operating.

4.5.8.2.1 Vibration. Vibration tests for both the operating and non-operating portions of the components flight shall be specified in the model specification. MIL-STD-810 shall take precedence if the specified test is more severe than the expected flight conditions.

4.5.8.2.2 Acceleration. An acceleration test for both operating and non-operating portions of the flight shall be specified in the model specification.

4.5.8.2.3 Temperature. High temperature, low temperature, and temperature shock tests for the operating and non-operating periods of the flight shall be specified in the model specification.

4.5.8.2.4 Altitude. An altitude test for the operating portions of the flight only shall be specified in the model specification. For components operating in the boost phase or in air vehicle propulsion systems, a rapid ascent test shall be performed in addition to determining performance at extreme altitude, including space.

4.5.8.3 Combined environmental tests. Combined environmental tests shall be specified in the model specification for those component operating environments that will be encountered during operation and whose interactions represent the most severe operating conditions.

4.5.8.4 Limit tests. Limit tests shall be specified in the model specification for the most severe stress producing environments to determine operating margins of safety. Limit tests shall be performed for each stress environment which affects the failure modes, as determined from the failure mode analysis.

4.5.9 Dissassembly and inspection. After completion of all qualification tests scheduled for a test article, it shall be disassembled for examination of parts to disclose excessively worn, distorted, or weakened parts or deterioration of functional capability.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. Propulsion system components shall be preserved and packaged in accordance with the applicable method or submethod of MIL-P-116.

5.2 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

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Part name

Name of manufacturer

Manufacturer's part number

Manufacturer's serial number

Contractor

Contractor's part number

Contractor's order number

Date of acceptance

6. NOTES

6.1 Intended use. The component system covered by this specification is intended for use in either manned or unmanned remote launched, air launched, and space craft vehicle systems. Propulsion feed system requirements are contained in MIL-P-27409.

6.2 Symbols. The symbols used herein and in the model specification will be those in American Association Standard ASA Y10.14-1959. Symbols not included in the preceeding standard that may be required shall be defined and included in the model specification.

6.3 Definitions.

6.3.1 Absolute filtration rating. The absolute filtration rating is a value where all particles greater than the rated size (based on a standard contaminate) will be removed by the filter.

6.3.2 Burst pressure. Burst pressure is the highest pressure to which a component may be subjected without catastrophic failure. Subsequent operation is not required.

6.3.3 Droop. Droop is the decrease in regulated pressure caused by a decrease in the reference force as the metering element opens from the closed to full open position.

6.3.4 Lockup. Lockup is the condition reached in pressure control components when the downstream pressure reaches a value where the component shuts off flow.

6.3.5 Nominal filtration rating. The nominal filtration rating is a value where 98 percent or more of the particles larger than a given size (based upon a standard contaminate) will be removed by the filter.

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6.3.6 Operating life. Operating life is the total number of duty cycles including Quality Conformance tests, but prior to overhaul or replacement of worn parts (except for specified limited life parts).

6.3.7 Parts list. A complete listing of the parts required to assemble a complete component of a specific configuration.

6.3.8 Procuring activity. Procuring activity is the activity which negotiates the contract.

6.3.9 Proof pressure. Proof pressure is the highest test pressure to which an item is subjected without leakage, deformation adversely affecting normal operation, or permanent set.

6.3.10 Reliability. Reliability is the probability that a component will perform a required function under specified conditions, without failure, for a specified period of time.

6.3.11 Service life. Service life is the sum of the storage life and operating lives.

6.3.12 Storage life. Storage life is the time that the component can be stored after Quality Conformance tests without replacement of parts and subsequently operate within specified limits. This includes both shelf type storage and storage after assembly in operating system.

6.3.13 Working pressure. Working pressure is the maximum expected operating pressure to which the component will be exposed, considering surge pressures and operating temperatures.

6.4 Applicability to NASA. This specification has been coordinated with the National Aeronautics and Space Administration, Washington, D.C. and approved for use by their activities when specifically invoked in a contract or order.

Custodians:

Air Force - 12

Review Activities:

Air Force - 12, 14, 19, 70

Preparing activity:

Air Force - 12

Civilian Agencies Interest:

NASA

Reviewer/user information is current as of the date of this document. For further coordination of changes to this document, draft circulation should be based on the information in the current DoD Index of Specifications and Standards.

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APPENDIX I

10. SCOPE

10.1 Scope. This appendix establishes the outline and instructions to be used by manufacturers in the preparation of a rocket propulsion fluid component model specification.

20. APPLICABLE DOCUMENTS

20.1 The following specification in effect on date of invitation for bids or request for proposal forms a part of this appendix to the extent specified herein:

Specification

MIL-C-27410	Components, Rocket Propulsion Fluid System, General Specification For

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. APPLICATION

30.1 A complete model specification in accordance with the outline and instructions for preparation as specified herein, shall be prepared for each specific rocket propulsion fluid component model. Changes to this model specification shall not be accomplished by means of amendment sheets, but must be incorporated in the body of the model specification prior to approval. Revisions by amendment form, to an approved model specification, which has been released and is forming a part of a contract, will be acceptable to the procuring activity. Each amendment shall include and supersede the previous amendment. Revisions shall be incorporated into the body of the specification. An appropriate approval status shall be shown on page 1 of the model specification.

30.2 The rocket propulsion fluid component manufacturer's model specification shall be a fully definitive specification and shall include all applicable standard requirements of MIL-C-27410 supplemented by the additional data requirements and statements shown herein. The paragraphs of MIL-C-27410 shall be repeated verbatim when applicable. When departures are necessary from the requirements of this appendix and of MIL-C-27410, the details of such departures shall be stated as specific requirements bearing the same section, paragraph, heading, and numbering as in the appendix and MIL-C-27410. Such departures shall be indicated by the symbol "Ø" preceding the paragraph number.

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30.3 Parenthetical sentences, phrases, and words are included herein for the guidance of the feed system manufacturer in connection with the preparation of the model specification. Parenthetical statements shall not be copied in the model specification.

30.4 The specification number shall be the number assigned by the component manufacturer. When revisions are made, they shall be designated by the use of a letter following the number, with a revision date therefore, which shall be shown on page 1. Only the specification number and the revision suffix letter, if applicable, shall be shown on subsequent pages. The first revision will be identified by the letter "A" and succeeding revisions will be indicated by other letters in alphabetical sequence except that the letters "I", "O", "Q", and "S" will not be used. Changes from previous revisions shall be indicated by vertical black lines in the left margin along the revised portion of the text. A summary of revisions shall be included in the model specification following the cover page.

30.5 For purposes of permitting preliminary evaluation of a proposed component design, or for release of approved rocket propulsion fluid component performance characteristics in connection with a vehicle design competition, the component manufacturer may submit a preliminary model specification. However, a complete approved model specification will be required for a contract. Preliminary model specifications shall be identified by the word "PRELIMINARY".

30.5.1 It is desired that the preliminary model specification be prepared in accordance with the requirements stipulated herein, except that it is recognized that certain information not available or pertinent to the purpose of the preliminary specification may be omitted.

40 FORMAT AND CONTENT

40.1 The form and description of the model specification follow.

(Preliminary, if applicable)

Approval Status (include applicable statement) (This draft has not been approved, DO NOT USE FOR PROCUREMENT PURPOSES) (Approved by the procuring activity letter _____)

(Specification No.) _____
(Date) _____
(a) Revised _____ (date)
(b) Revised _____ (date)

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(Preliminary Flight Rating or Qualification or
Preliminary Flight Rating and Qualification, as Applicable)

MODEL SPECIFICATION
COMPONENT, ROCKET PROPULSION FLUID:_____
(Insert Government designation if assigned)

(NAME OF CONTRACTOR)
(Contractor's Model Designation)

1. SCOPE

1.1 Scope. This specification covers the requirements for rocket propulsion fluid system components and includes the applicable standard requirements in MIL-C-27410 to the extent specified herein.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein. (List all documents referenced herein. This listing shall include the document number, revision letter, amendment number, date, and the complete document title. Titles shall be checked with the publication itself, rather than an index. SPECIFICATIONS, STANDARDS, DRAWINGS, and other PUBLICATIONS shall be listed numerically under these headings and in individual groups such as Federal or Military. No documents shall be listed in this section except those which are referenced in the body (Section 3 through 5 only) of the model specification wherein the extent of applicability shall be clearly indicated in those cases where the entire specification is not applicable.)

2.2 The following non-government documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein. (List all documents referenced herein. This listing shall include the document number, revision letter, amendment number, date and complete document title. Documents shall not be listed in this section except those referenced in the body (Section 3 through 5 only) of the model specification wherein the extent of applicability shall be clearly indicated in those cases where the entire document is not applicable.)

3. REQUIREMENTS

3.1 General.

3.1.1 Model specification. Components shall be designed, fabricated and tested as specified herein.

3.1.2 Pre-production. (Repeat text from MIL-C-27410.)

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3.1.3 Changes. (Repeat text from MIL-C-27410.)

3.2 Materials and processes. (Repeat text from MIL-C-27410.)

3.2.1 Metals. (Repeat text from MIL-C-27410.)

3.2.2 Rubbers, plastics, and elastomers. (Repeat text from MIL-C-27410.)

3.2.3 Lubricants. (Repeat text from MIL-C-27410.)

3.2.4 Finishes. (Specify all finishes, surface treatment, platings, coatings, or paints.)

3.3 Performance characteristics. (Specify the performance characteristics and include the intended application and system interactions.)

3.3.1 Operating attitude. (Specify the operating attitudes and performance characteristics of the components when operated in any attitude the units may be exposed to during their service life.)

3.3.2 Life characteristics. (Specify the component operating and storage life characteristics.)

3.3.2.1 Storage life. (Repeat text from MIL-C-27410.)

3.3.2.2 Operating life. (Specify the operating life to include acceptance testing and system checkout tests, and propulsion system operation.)

3.3.3 Leakage. (Specify maximum leakage for each component, source of leakage, fluids and leakage rates and describe provisions for disposition.)

3.3.4 Flow characteristics. (Specify flow characteristics to include flow rate versus pressure drop requirements for various percentages of maximum flow when required or various percentages of maximum flow position of the control element.)

3.3.5 Actuation. (Specify the method of actuation, response characteristics open and closed, repeatability of the response characteristics and the control mode; i.e., digital, proportional, etc.)

3.3.6 Power. (Specify the maximum power allowable for component operation with the tolerances and characteristics of the power supplied.)

3.3.7 Stability characteristics. (Specify the stability characteristics to include overshoot, undershoot, damping and dead band condition.)

3.3.8 Filtration. (Specify nominal and absolute filtration along with the dirt holding capacity of all filter elements.)

3.3.9 Draining. (Specify maximum spillage which may occur in disconnect operations.)

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3.4 Design. (Repeat text from MIL-C-27410.)

3.4.1 Misalignment. (Specify maximum allowable loads on the component parts and other attachments caused by axial, lateral, and angular misalignments in the tubing or the support structure under the most severe environmental condition including any deflection or misalignment accomodation features.)

3.4.2 Standards. (Repeat text from MIL-C-27410.)

3.4.3 Special tools. (Repeat text from MIL-C-27410.)

3.4.4 Pressure. (Specify working proof and burst pressure. The proof and burst pressures shall be at least 1.5 and 2 times the working pressure respectively.)

3.4.5 Installation characteristics. (Specify mounting details, overall dimensions, location, size and type of plumbing and electrical connections, servicing clearances and flow directions.)

3.4.6 Environment. (Specify the environments the component will experience in flight; i.e., vibration, acceleration, temperature, and altitude, as well as the non-operating environments; i.e. shipping, vibration and shock, the natural environments of the launch and test sites, and the launch environments for upper-stage systems component.)

3.4.7 Contamination. (Specify contamination levels and quantities of contaminants and each particle size level which the component must be capable of accepting without degradation in performance.)

3.4.8 Fluids. (Repeat text from MIL-C-27410.)

3.5 Construction.

3.5.1 Threaded connections and attachments.

3.5.1.1 Locking provisions. (Repeat text from MIL-C-27410.)

3.5.1.2 Pipe threads. (Repeat text from MIL-C-27410.)

3.5.1.3 Threads. (Repeat text from MIL-C-27410.)

3.5.2 Electrical devices. (Repeat text from MIL-C-27410.)

3.5.3 Hydraulic devices. (Repeat text from MIL-C-27410.)

3.6 Reliability. (Repeat text from MIL-C-27410.)

3.7 Interchangeability. (Repeat text from MIL-C-27410.)

3.8 Accessibility. (Repeat text from MIL-C-27410.)

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3.9 Identifying products. (Repeat text from MIL-C-27410.)

3.10 Nameplate. (Repeat text from MIL-C-27410.)

3.11 Drawings and diagrams. (Repeat text from MIL-C-27410.)

3.12 Parts list. (Repeat text from MIL-C-27410.)

3.13 Contamination control. (Repeat text from MIL-C-27410.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. (Repeat text from MIL-C-27410.)

4.2 Classification of tests. (Repeat text from MIL-C-27410.)

4.2.1 Development tests. (Repeat text from MIL-C-27410.)

4.2.2 Qualification tests.

4.2.2.1 Sampling instructions. (Repeat text from MIL-C-27410.)

4.2.2.2 Tests. The qualification of propulsion system components shall consist of tests (4.5) which are applicable to a particular component. Additional tests or more severe requirements may be specified herein. Specify all qualification tests in detail, to include the following

a. Number of components

b. Testing conditions

c. Criteria for success or failure.

4.2.3 Quality Conformance tests. (Repeat text from MIL-C-27410.)

4.3 Failures and retest of test samples.

4.3.1 Development tests. (Repeat text from MIL-C-27410.

4.3.2 Qualification tests. (Repeat text from MIL-C-27410.)

4.3.3 Quality Conformance tests. (Repeat text from MIL-C-27410.)

4.4 Test conditions. (Repeat text from MIL-C-27410.)

4.4.1 Cleaning. (Repeat text from MIL-C-27410.)

4.4.2 Temperature and pressure. (Repeat text from MIL-C-27410.)

4.4.3 Attitude. (Repeat text from MIL-C-27410.)

4.4.4 Test fluid. The test fluid for each tests shall be _____.

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4.5 Test and test mehods. List and describe in detail the following:

a. Tests

b. Test procedure

c. Test system design

The test system for the qualifications shall have the same dynamic characteristics as the system in which the component is intended. Required instrumentation and allowable measurements shall be specified herein.

4.5.1 Examination. (Repeat text from MIL-C-27410.)

4.5.2 Strength.

4.5.2.1 Proof pressure. (Specify proof pressure as applied at all ports simultaneously for three (3) minutes.)

4.5.2.2 Burst pressure. (The component shall have the burst pressure, as specified herein, applied and maintained for a period of not less than three (3) minutes.) The method of applying the pressure shall be specified herein.

4.5.2.3 Surge pressure. (Specify the test system procedure and requirements of this test.)

4.5.2.4 Mechanical load. (Repeat text from MIL-C-27410.)

4.5.3 Leakage. (Repeat text from MIL-C-27410.)

4.5.3.1 Internal. (Repeat text from MIL-C-27410.)

4.5.3.2 External. (Repeat text from MIL-C-27410.)

4.5.4 Functional.

4.5.4.1 Pressure drop. (Repeat text from MIL-C-27410.)

4.5.4.2 Pressure control. (Specify in detail the test procedure and test configuration.)

4.5.4.3 Flow control. (Repeat text from MIL-C-27410.)

4.5.4.4 Response. (Specify, in detail, this test conducted over the range of conditions covering the design capacity of the units.)

4.5.4.5 Power requirements. (Repeat text ffom MIL-C-27410.)

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4.5.5 Endurance. (Specify the endurance test.) (This test, as a maximum, shall be three times the operating life requirements specified in 3.3.2.2.)

4.5.6 Electrical system.

4.5.6.1 Explosion-proof. (Repeat text from MIL-C-27410.)

4.5.6.2 High ambient air temperature. (Repeat text from MIL-C-27410.)

4.5.6.3 Dielectric strength. (Repeat text from MIL-C-27410.)

4.5.6.4 Radio interference. (Specify the detailed test procedure for each particular component.)

4.5.7 Fluid compatibility.

4.5.7.1 Internal. (Repeat text from MIL-C-27410)

4.5.7.2 External. (Repeat text from MIL-C-27410.)

4.5.8 Environment.

4.5.8.1 Ground handling and storage.

4.5.8.1.1 Mechanical shock. (Specify the procedure for shock test.)

4.5.8.1.2 High temperature. (Repeat text from MIL-C-27410.)

4.5.8.1.3 Low temperature (Repeat text from MIL-C-27410.)

4.5.8.1.4 Temperature shock. (Repeat text from MIL-C-27410.)

4.5.8.1.5 Humidity (Repeat text from MIL-C-27410.)

4.5.8.1.6 Salt fog. (Repeat text from MIL-C-27410.)

4.5.8.1.7 Sand and dust. (Repeat text from MIL-C-27410.)

4.5.8.2 Flight environment. (Repeat text from MIL-C-27410.)

4.5.8.2.1 Vibration. (Specify the vibration tests for the operating and non-operating portions of the components flight.)

4.5.8.2.2 Acceleration. (Specify an acceleration test for the operating and non-operating portions of the flight.)

4.5.8.2.3 Temperature. (Specify the high temperature, low temperature, and temperature shock tests for both the operating and non-operating periods of the flight.)

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4.5.8.2.4 Altitude. (Specify the altitude test for the operating portions of the flight only.)

4.5.8.3 Combined environmental tests. (Specify the combined environmental tests for those component operating environments that will be encountered during operation and whose interactions represent the most severe operating conditions.)

4.5.8.4 Limit tests. (Specify limit tests for the most severe stress producing environments to determine operating margins of safety.)

4.5.9 Disassembly and inspection. (Repeat text from MIL-C-27410.)

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. (Repeat text from MIL-C-27410.)

5.2 Marking. (Repeat text from MIL-C-27410.)

6. NOTES

6.1 Intended use. (Repeat text from MIL-C-27410.)

6.2 Symbols. (The symbols used herein will be in accordance with American Association Standard ASA Y10.14-1959 and the following: If symbols not included in this standard are used, a tabular list of the symbol(s) and its definition(s) will be included under this paragraph.)

6.3 Definitions. (Repeat text of all subparagraphs of this paragraph in the order presented adding any definitions needed for undefined new terms used in tabular form under the succeeding paragraph number.)

(Paragraph 6.4 of MIL-C-27410 is pertinent to this military specification only and is not to be included in the component manufacturer's model specification.)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863 WHICH IS OBSOLETE C-8270

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